DCG Student Assignment - Desk light

EXAMINATION NUMBER: 141716

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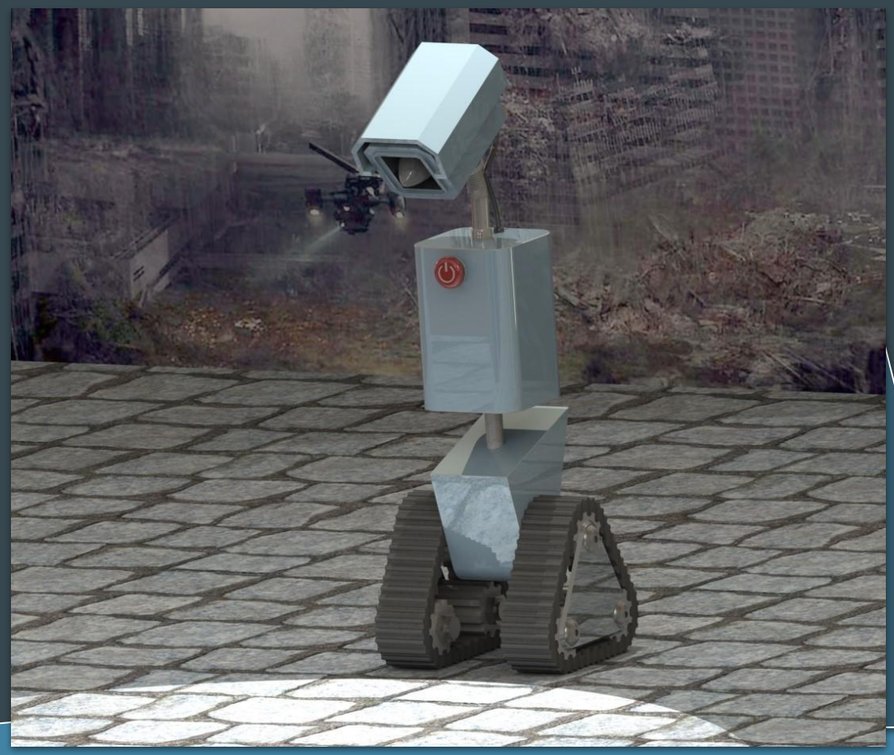


Desk Light

Design Research

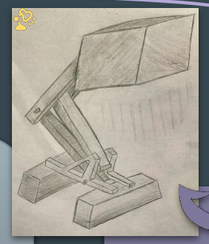
- My sketches

* - Secondary research

Noun: Desk lights are **objects** that provide **focussed illumination** on a **desk** to **promote task work** such as reading or studying.

Target Market

The target market is **very important** as the **design** , **style** and **functionalities** of the product **changes** depending **on who it's aimed at** . There are **several types of people** that would **buy** desk lights: Students, remote workers, gamers, readers and professionals. Each demographic has their own set of requirements for a desk light so the target market is very important.



Moveable neck

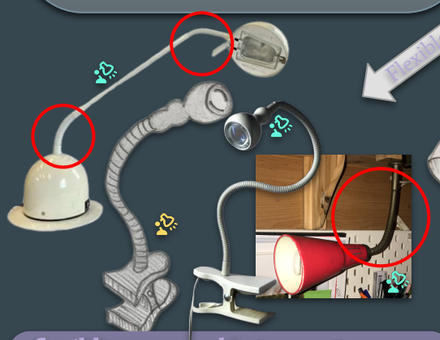
Moveable Neck

A moveable neck allows you to **direct** the **light precisely** where you need it, making it an essential feature for customization in your workspace. The ability to angle and adjust the light ensures you have optimal lighting condition for any task work at hand.

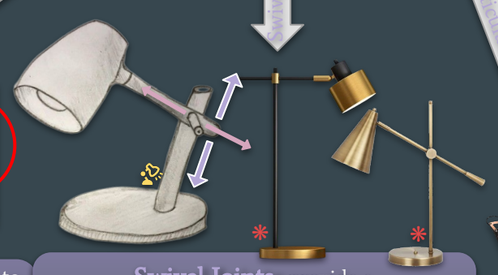
Flexible Gooseneck

Swivel Joints

Articulated arms



flexible gooseneck designs allow you to bend and twist the neck into different positions. This design is versatile and can be adjusted to various angles and heights.



Swivel Joints provide a vertical/horizontal pivoting motion to focus the illumination in the desired spot.

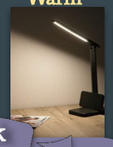
articulated arms consist of multiple segments with hinges to be extended bent and rotated in order to position the focused illumination wherever you please.

When researching all the different types of necks i realised these three were the most commonly used and most practical as well. However, amongst the three i think the articulated arm is the best design because of its wide manoeuvrability. Swivel joints and goosenecks are limited to their fixed state even while extended whilst articulated arms have multiple segments so they can be extended to their full length to cover a wider area.

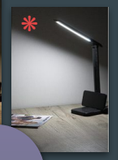
Adjustability

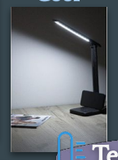
For maximum efficiency and purpose, desk lights **must be able** to adjust to the **consumers preference.** Adjustable settings on a desk light may include brightness , temperature and a moveable neck .

Temperature



Warm

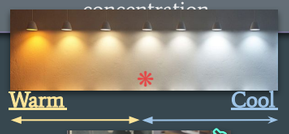




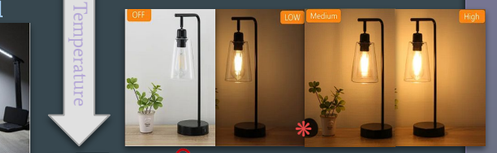
Cool

Temperature

The temperature colour emitted from a desk light can **control the ambiance** that is to be portrayed in your workspace. **Warmer temperatures** create a **cozy** , **inviting** atmosphere that's ideal for reading or relaxing, while cooler **temperatures** mimic **natural daylight** and are better suited for tasks requiring focus and concentration.



Warm Cool

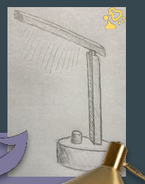


We can see by the graphic shown above that the level of brightness can create different moods of atmosphere.

brightness

Brightness

The brightness of a desk light is a **crucial factor** to consider when evaluating its adjustability . The brightness directly **impacts any task work** such as reading, studying or working and being able to control the brightness to suit your personal need is ideal. **Brighter** settings can **help reduce eye strain** during tasks that require focus and concentration, while **lower** brightness options create a more **relaxed atmosphere** for casual activities or **ambient** lighting.



Aesthetic

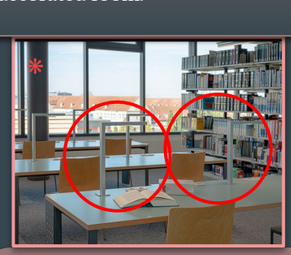
The aesthetic of a desk light is often **influenced** by the target market . With quite a stable market, **brands** are required to **innovate different** type of **designs** in order for their desk lights to **stand out** . This allows **everyone** to **find** the **perfect desk light** suited to their own **personal style** and **preference** . Desk light aesthetics range from **simple and minimal** all the way to **flashy and futuristic** with the **size varying** constantly and the materials **shifting** accordingly to provide **different** aesthetics for everyone. Some **brands** may only produce metal desk lights to indicate **quality and luxury** whilst other brands may produce hard plastic based desk lights to **cut the cost** but **still provide** an **aesthetically pleasing** desk light.

Location

A desk light's location / environment is often determined by its aesthetic. When placing a desk light it should compliment its surroundings. For example, a modern, sleek desk light may look out of place in a traditionally decorated room.

Office:

Long extendable desk light on a large desk. A small, portable desk light would be a good example of a desk light not ideal for this location as it would not be able to provide the coverage needed.



Library:

Sleek design to provide minimal and decluttered area. It's also screwed in to prevent vandalism and the taking of these desk lights.

Home work space:

Small clip on desk light on a small desk to provide more space. An example of a desk light that **wouldn't suit** this area would be one that would wouldn't be clipped on but on the desk instead with a base.

wouldn't suit

Light Bulbs

LED - Light emitting diode

- These are **semiconductor devices** that **emit light** when electric current is passed through it. These bulbs tend to have **longer life spans** than filament bulbs as you do not have to replace it as a often.

They provide **less carbon emissions** which make them **more sustainable** for everyday use however most led light produce **blue light** which **disrupts** the consumers **circadian rhythm** when using these bulbs at night which can lead to **bad sleeping habits.**

Led - Light emitting diode

Diffuser which allows the light emitted from led module to spread evenly.

Led module that emits light.

Thermal transfer pad and heat sink to manage and dissipate the heat generated by the LED chips.

Cap of the bulb helps secure the bulb into its fitting.

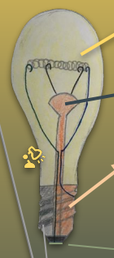
Electrical contact to power the bulb.

Incandescent bulb (filament)

- These traditional light bulbs emit light when an **electric current** flows through a **thin filament** , **heating** it to a high temperature **until** it **glows** and emits visible light.

Although these lights may be favoured by those who like the traditional aesthetic look, they have a **great amount of drawbacks** . Filament bulbs **require more power to work** and at that they still are a lot **less efficient.** Much of the electricity being used is turned into heat rather than just light and **disposing** these bulbs can be **hazardous** due to the use of tungsten and lead to make them.

Incandescent bulb (filament)



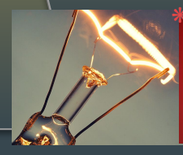
Filament that heats up and eventually turns red to produce light.

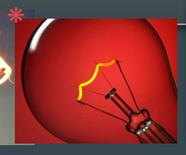
Stem of the bulb allows the electricity to flow without gas or air leaks.

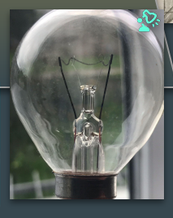
Vitrite which is very hard and difficult to fuse is used in filament bulbs as an insulator which is crucial to prevent the bulb from melting.

Electrical contact to power the bulb.

Cap of the bulb helps secure the bulb into its fitting.







Controls

The controls of a desk light have significant importance when defining functionality and user experience. These controls come in various forms and allows users to adjust the brightness levels, choose the color temperature, toggle the light on or off with ease and do extra functionalities provided there are any.

Control built into the cord to make a simple Inline control for easy on/off switch.

Dimmer inline switch built into cord for precise brightness control.

Controls built into desk light base , neck or head. Users don't have to search for an inline control on the cord and can look more seamless and aesthetically pleasing.

Remote control to access desk light wirelessly and at a distance. Often used in big spaces with farther away desk light.

My Image My Sketches Secondary research

Material

The **choice of** material plays a significant role in **determining the** cost of a desk light. Desk lights can be made from various materials, ranging from inexpensive **plastics** to high-end **metals**. Materials like premium **metals** and **wood** may drive up the cost due to their **quality**, aesthetic appeal, and **longevity**. In contrast, options like plastic may result in more affordable desk lights but might compromise on durability and overall quality. This indicates that the material choice strongly influences the price range of a desk light, with options available to suit various budgets and preferences in mind.

Material

Cost

The cost of a desk light can be determined by a numerous amount of things. However, there are **three stand-out factors** that **often always influences** the **cost**: The **Material**, **Brand** and **Extra Functionalities**.

Brand

Bran d

Brands wield a **powerful influence** on the cost of desk lights. If a brand is **well established** and is **reputable**, they can **command** a **higher price** for their items as people would already associate it with reliability and quality. Brands like these tend to sell a lot more than smaller ones which can even drive the price farther up as it creates that demand however brands like ikea like to stick to **affordable prices** whilst **maintaining quality** as much as possible and with that they gain a good **reputation**. Brands like **dyson** have produced great products in and out of desk lights which are typically good quality and with that **reputation** they use it to **scale** their **price up** for their desk lights **significantly**.

plastics

Plastic

Plastic desk lights are the **best option** to pick when looking explicitly for **something cheap** but still **good quality**. Plastic desk lights still have the properties needed to function accordingly but above all else it is the aesthetic that **is affected**. The plastics **texture / finish** can make or break the aesthetic and either rise or lower the cost of it. Upon my research i found that more people would be inclined to buy a plastic desk light if it had a matte finish as opposed to a glossy finish.

Metal

When a metal is in pristine condition its **properties** are **much sought out for** in the manufacturing process of desk light. These properties include **conductivity**, **malleability**, and **corrosion resistance**. These attributes are highly ideal for something that might be used daily such as a desk light. Not only do metal desk lights have great properties, they also provide for a sleek aesthetic which is why it is often picked a primary material for desk lights.

Wood

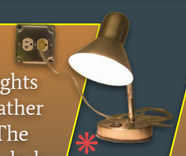
Similarly to metal, Wood is a much sought out material for those looking to buy an aesthetically pleasing desk light. This immediately drives up the cost of the desk light as it is **more in demand**. This is not the only reason for the higher price. Wood may have a much **longer manufacturing process** depending if the desk light has been **handcrafted**. People may also pay **higher prices** for a **good looking grain** on the wood for their desk light. Even if the desk light **had not been handcrafted**, the manufacturing process is **still long** because **unlike** plastic and metals, Wood **cannot** be **molded** into a **shape** to provide an immediate result so other **added machinery is used** to get the same outcome.

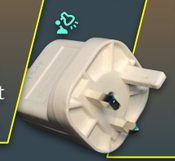
Power source

Desk lights are either **plugged in** or **battery powered** with both having their advantages and disadvantages. Battery powered desk lights have a slight edge in terms of portability and being able to be utilised in power outages but personally i would much rather have a plugged in desk light as i do not want to have to think about regular charging but for those that are on the go and might require a portable desk light, battery powered is the best option.

Plugged in:

Most traditional desk lights tend to be plugged in rather than battery powered. The power fed into into the desk light is constant and will stay the same brightness whilst plugged in as opposed to battery powered where the amount of light may vary depending on the amount of battery left to power it.



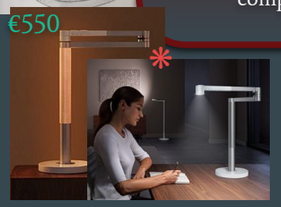


Battery powered:

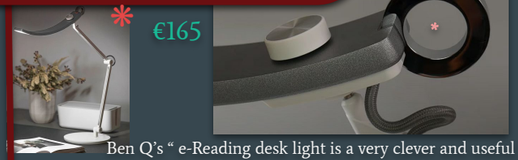
Some battery powered desk lights have solar cells which can make them much for favourable in terms of sustainability while on the other hand aa-alkaline batteries can produce harmful chemicals when decomposing making them harmful for our environment

Extra Functionalities

For those who will be staying at their desks for longer time periods, it would be **ideal** to have a **multi purpose** desk light with extra functionalities rather than one that solely emits light. Companies add in these extra functionalities for a more **ease of life** and better practicality but it comes at an added price as compared to a standard desk light.



Dysons “solar cycle morph” has a pivot type arm and its distinctive extra functionality is its ability to turb back into a desk lamp by repivoting its arm back into its stem which then diffuses the light via its meshed texture which is quite clever but in my opinion i do not believe this feature alone is enough to raise the price to what it is.



Ben Q’s “ e-Reading desk light is a very clever and useful lamp. It is specifically designed for people who sit at their desks for long periods of time. Its main distinctive feature is its automatic light detection with the ring* changing the light temperature and brightness automatically upon touch. Another clever extra functionality is the design of the lamp shade itself. It is anti-glare and is curved in a u shape to attempt to cover as much desk space as possible.

Reflection

In my opinion the Desk light market offers a lot of diverse designs suited for its unique users. There's a lot to digest from their individual shapes to their contrasting functionalities. From this research I have discovered many aspects of the desk light i hadn't noticed before such as how some factors like the aesthetic determines so much other factors such as material, location and even light bulbs. I also learnt the advantages and disadvantages of the materials used to make desk lights and how innovative extra functionalities can drive the price way up. I personally think that desklights should be incorporated into all work spaces to help those that work for long periods. Having a desk light can provide a lot of advantages for the consumer and will in long term provide for more productive task work.

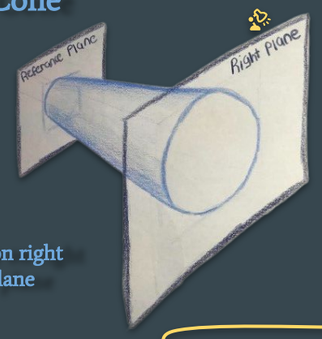
Identifying the geometry

Identifying the geometry

Identifying the geometry

Identifying the geometry

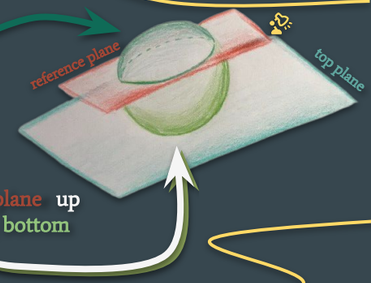
Truncated Cone



Surface Lofted base from circle on right plane to circle on reference plane

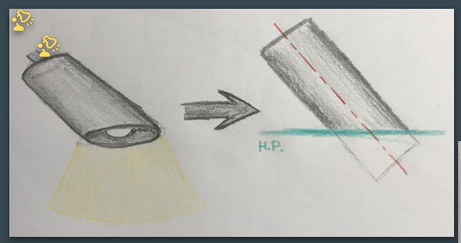
Circle on top plane

Extrusion



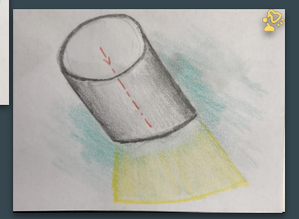
Sphere on top plane, reference plane up and split, Consume cut bodies bottom sphere.

Prisms on right plane



Oblique cylinder cut by top plane

Cut surface is elliptical

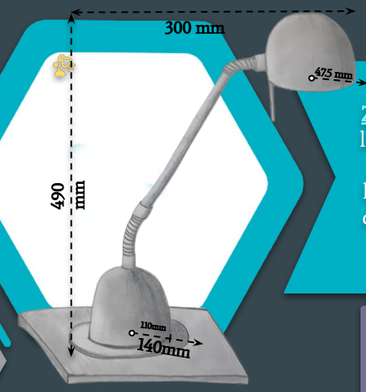


Desk Light

Compare & Contrast

My Image Secondary My Sketches

Both desk lights certainly differ in aesthetic and ergonomics. There is a distinctive look to each of the desk lights which allows us to differentiate between the two easily but we will also compare all the other aspects of the two desk lights which might not be as easy to spot.



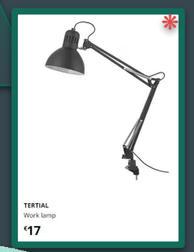
Cost

Upon research i figured out that the **difference in cost** of these two desk lights had quite a **significant gap** which surprised me. As expected the Ikea desk light came out **relatively cheap at** €17 which is quite good for what you are getting in comparison to the many desk lights already produced. However **in contrast to the** Zippo desk light it is **favourably cheaper** as it comes out at €62.37 which is certainly a **steeper cost.**

Ikea have always had a **formidable reputation** for **quality products** at a **low price.** They have been constantly **dominating** the furniture market and without a doubt the **desk light market** as well.

It is also important to take into account that the Zippo desk light was purchased in the far past which might have influenced its high cost.

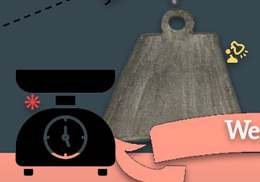
Material



Material

The Ikea desk light is suited with **thin sheet steel** that contains an **epoxy/polyester powder coating** on the reflective shade and as well as the arm. With the Ikea desk light being overly made of metal, the Zippo desk light is **fixed with hard plastic** to suffice for a strong sturdy build. **Both desk lights** apply for having a **firm, durable build** whilst having used **two very different materials** but there are **still advantages and disadvantages** of using these materials.

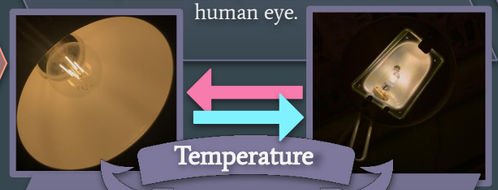
Weight



Both desk lights also differ heavily in weight. The Zippo desk light is considered heavy for a desk light. It weighs approximately **2.5kg** as opposed to Ikeas **1kg** desk light which is much taller and longer however this **weight has a purpose.** Most of the weight is built up in **the base** of the Zippo desk light as it rests on the table. This allows the desk light to remain in place whilst the user adjusts the neck to their preference. The Ikea desk light is **clamped** and fixed to a position on the side of a table so it does not need to worry about moving around unwantedly.

Brightness

Both desk lights are **not equipped** with **brightness control** and hence output the **same level** of brightness. The Ikea desk light holds a max output of **60 watts** and the Zippo outputs **50 watts** . This difference is tiny and is barely noticeable to the human eye.



Temperature

Although **both** desk lights emit a **Warm temperature** they both have different light bulbs. Led lights **can emit** cold temperatures **but** in the Ikea desk light **they chose** a **warm** temperature. In the zippo desk light, it is kitted with an incandescent bulb which **strictly** only emits **warm** temperatures.

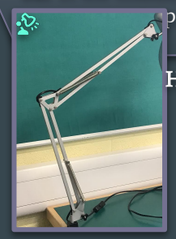
Zippo is a **discontinued** desk light manufacturing company. It is a more **traditional** desk light and was manufactured a distant time ago as seen by its aesthetic .

Adjustability

Both desk lights meet the standards of adjustability to suit the customers preference. They both have **different moveable necks** but in terms of **temperature** and **brightness** they are both **similar** .

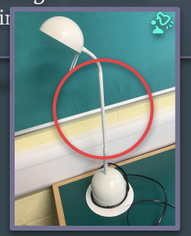
Moveable Neck

The neck on the Ikea desk light is far more complex than the neck on the Zippo desk light. It has an articulated arm which has a **significant amount** of **segments and hinges** to allow it to bend and stretch to the consumers preference. Not only does it have more segments but it also has an **overall longer and taller** structure **than the** Zippo desk light which has a short flexible gooseneck styled design. The gooseneck on the Zippo light is not prominent throughout the whole neck. It has a **Hard plastic solid bar** in the middle to give the light a stronger structure to prevent it from droopi



Hard plastic solid bar

Articulated Arm



Extra Functionalities

When it comes to extra functionality, the main focus is Ikeas desk light rather than Zippos. It features a resting area for the desk light that can either be **desk mounted or wall mounted** where as the Zippo desk light is solely grounded. It has small screw holes on the resting area so you can drill the clamp into the wall at any place you desire and will remain in place in a sturdy fashion.

Hard Plastic

As stated before, **Hard plastics** have a strong , firm build but they might not have the same aesthetic appeal as other materials as it comes out with a glossy finish often times as seen on the Zippo , **compared to** the sleek result of steel in the Ikea desk light. Hard plastics also resist corrosion and are resistant to many chemicals but it is generally not as heat-resistant as some other materials such as steel in the Ikea desk light.

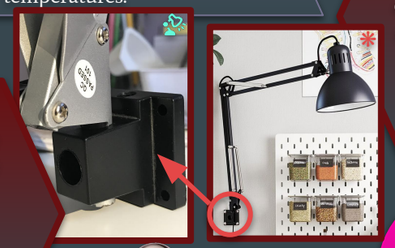
steel with epoxy/polyester powder coating

Steel is a popular choice of metal in the manufacturing of desk lights as it is sturdy, durable and can **withstand wear and tear.** It is also notable for having a **high heat resistance.** It's able to do this because it is **tough** and **quite ductile** so no matter the load or sudden impact on the desk light it will **always be able to maintain its shape** . However, steel is made of iron and carbon which means it is **prone to rust** but **due to** the Ikea desk light's **epoxy/polyester powder coating** , it can **withstand rust/corrosion** whilst maintaining a **sleek design.**

Controls

The controls displayed on the two desk lights come in the form of an **inline control** switch on the Ikea desk light and the controls **built into the desk lights base** in the Zippo . Both desk lights feature an **on and off switch,** they are just placed in two **different locations.**

Due to Zippos desk light being incandescent bulb based, it becomes quite hot quite fast. This means the shade of the desk light would be hard to maneuver by the customer. They tackled this issue by putting a handle on the bottom of the shade. It has a great purpose but it also just works well as an easy to access handle for focussing illumination



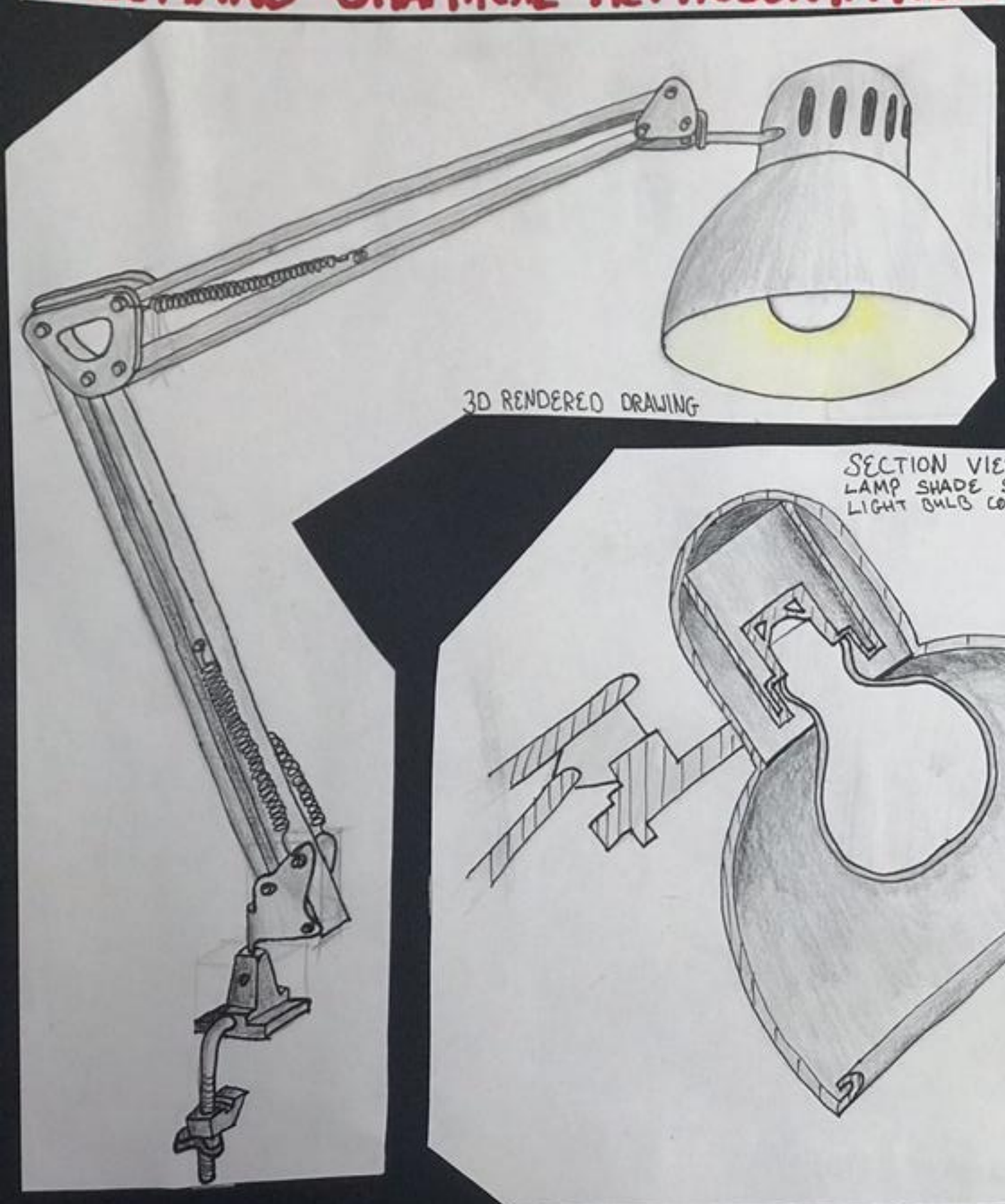
Light Bulbs

Desk Lights like the Zippo are not very seen anymore due to most desk light companies moving to the more **sustainable option** of **led bulbs** . However companies are **also using Led filament bulbs** to **retain** the aesthetic of filament whilst also maintaining the **efficiency** of **Led bulbs** as seen in the Ikeas desk light

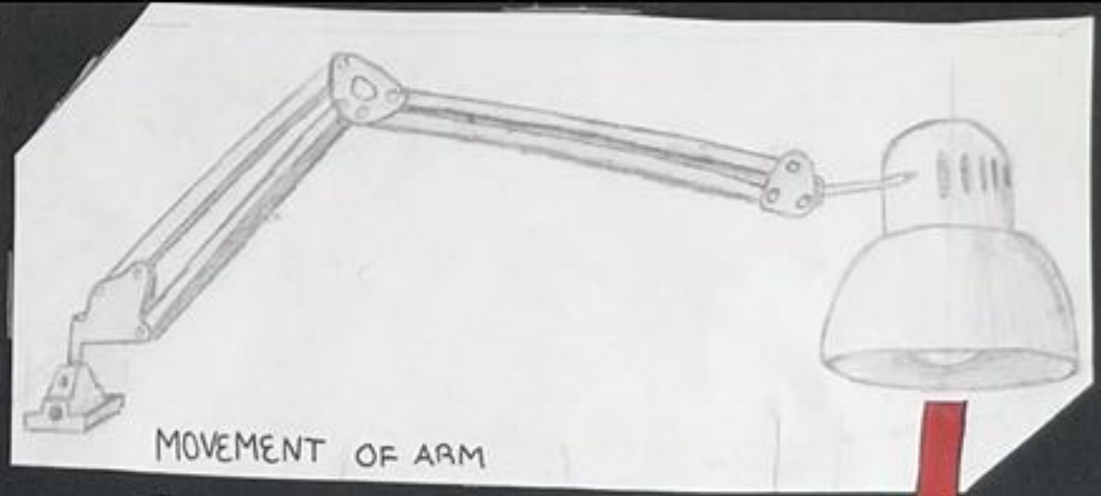
Reflection

After comparing these two products I have become more knowledgeable about the design and benefits they each have. I am now more aware that although the desk lights are different they both fulfill the standards of a desk light. I learned that the dates that these two desk lights were manufactured heavily affected the difference between them as the zippo desk light is something more traditional and classic whilst the ikea desk light is a lot more modern which gives the two desk lights a strong contrast against each other. Overall, the Ikea desk light is certainly better than the zippo desk light in terms of practically everything. It is a lot more sustainable which is highly regarded in the world today but the zippo desk light is also much smaller which may be preferable to those in more confined spaces. I believe that all my research from now will help me throughout this project as i now have a broad understanding of a desk lights physical form and shape in particular, the Ikea desk light and the zippo desk light.

FREEHAND GRAPHICAL REPRESENTATION

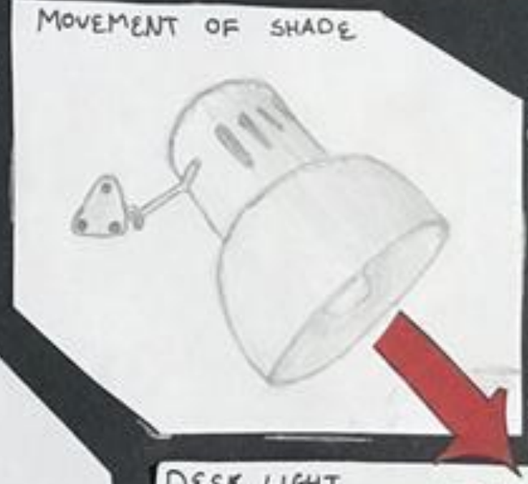


3D RENDERED DRAWING



MOVEMENT OF ARM

3D VIEWS

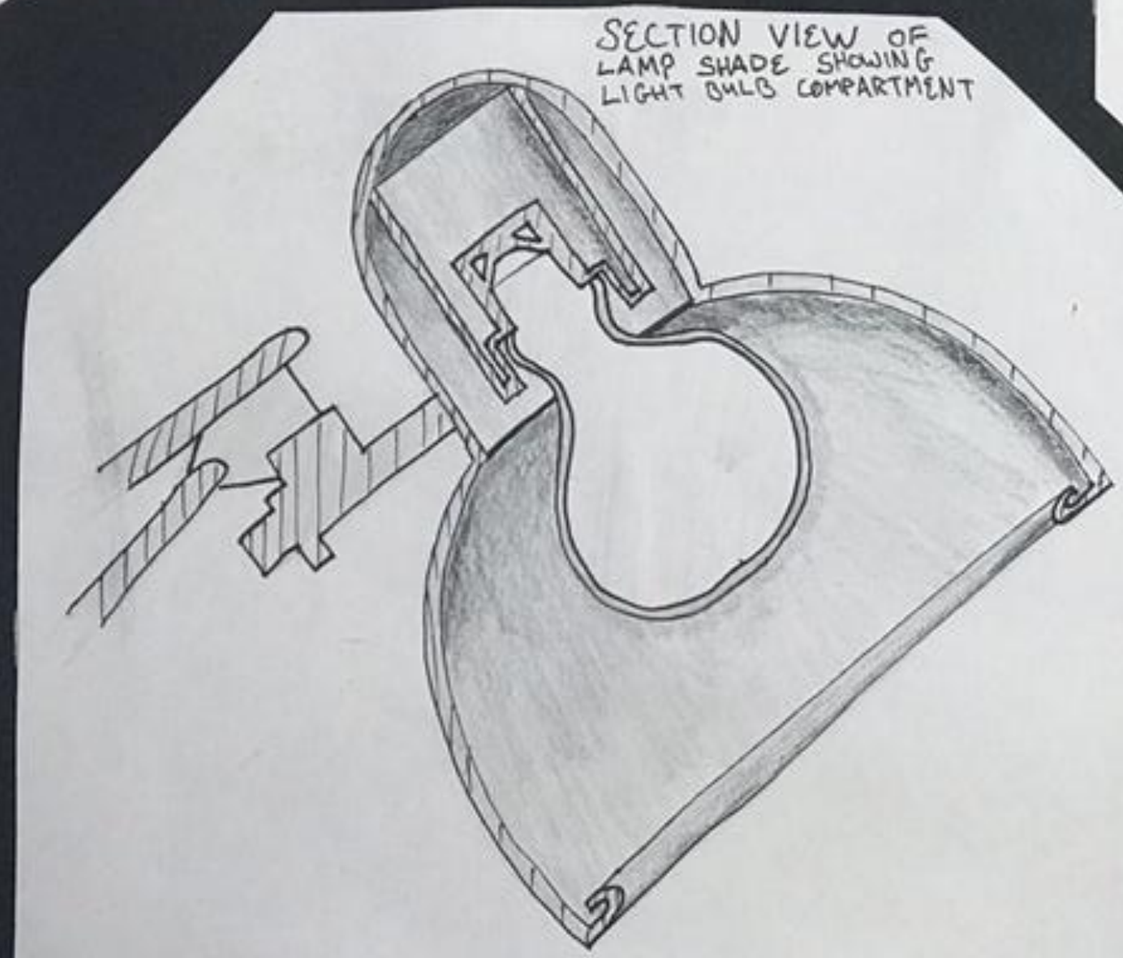


MOVEMENT OF SHADE

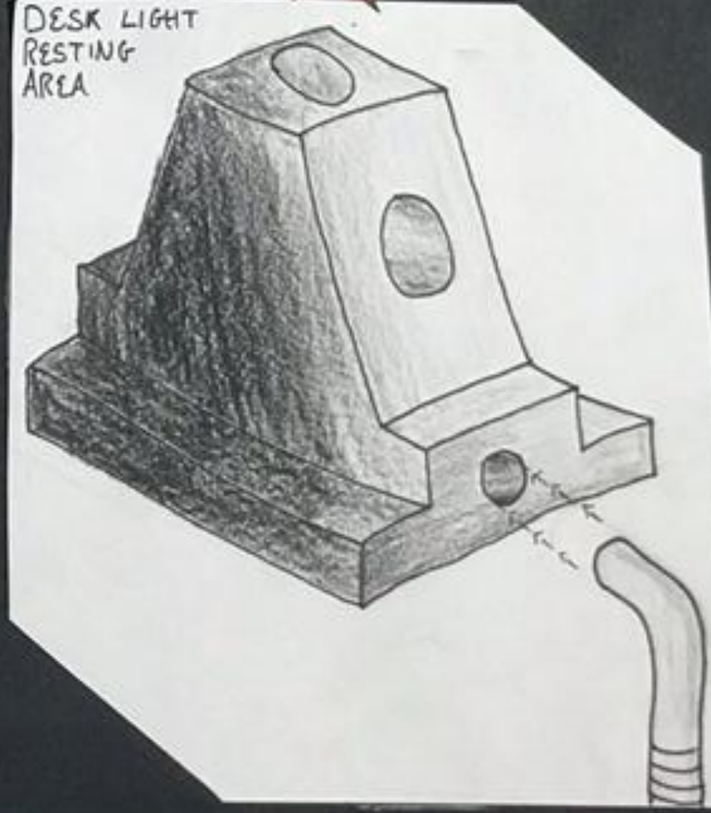
DIRECTION OF LIGHT BEFORE MOVEMENT

DIRECTION OF LIGHT AFTER MOVEMENT

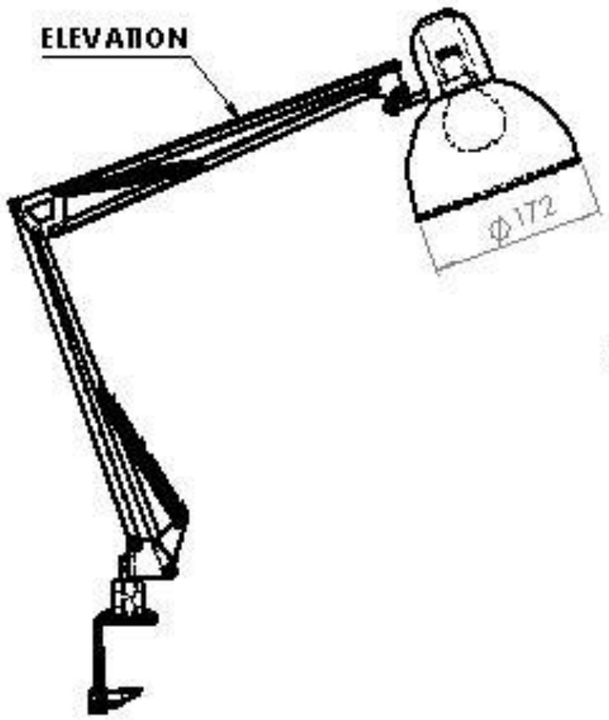
SECTION VIEW OF LAMP SHADE SHOWING LIGHT BULB COMPARTMENT



DESK LIGHT RESTING AREA

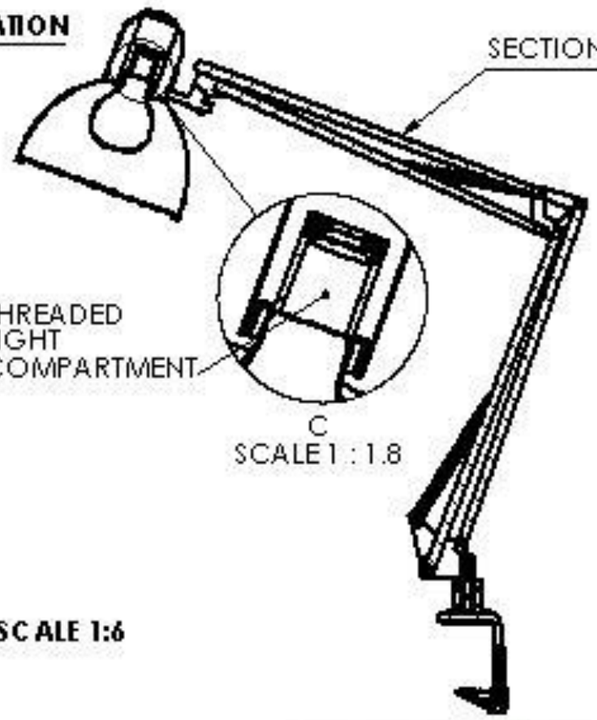


ELEVATION



END ELEVATION

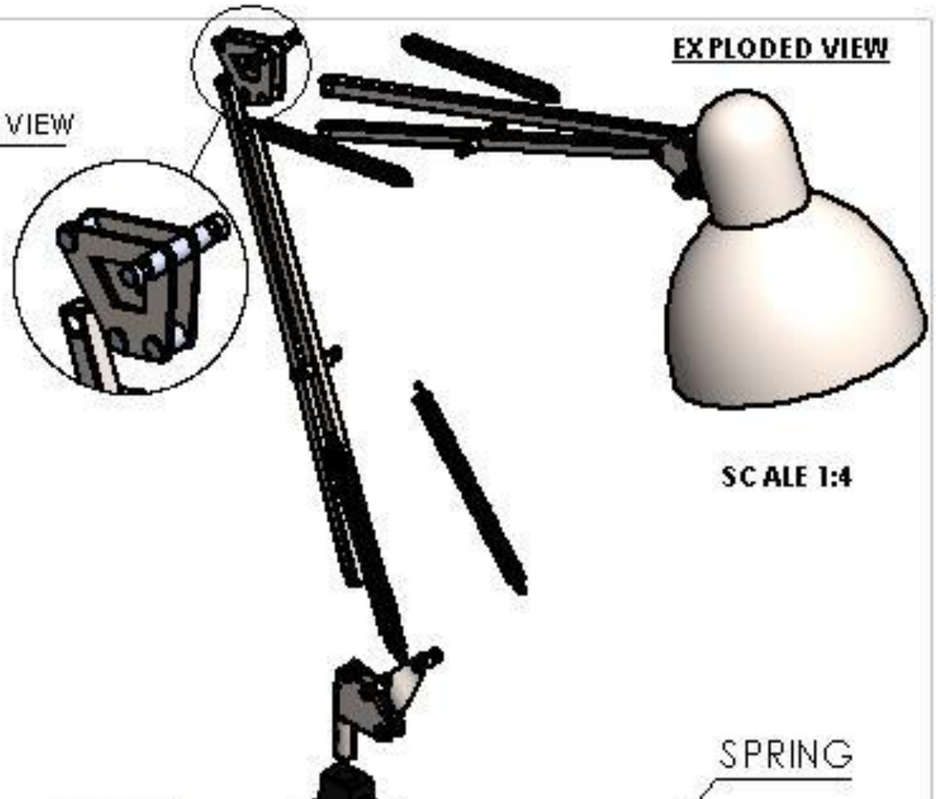
SECTION VIEW



C
SCALE 1 : 1.8

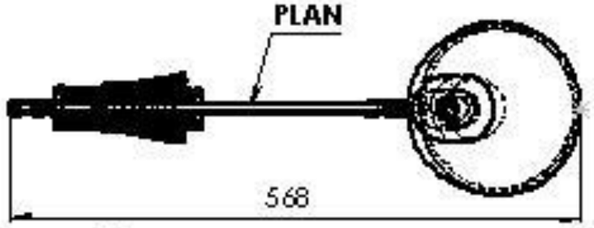
SCALE 1:6

EXPLODED VIEW

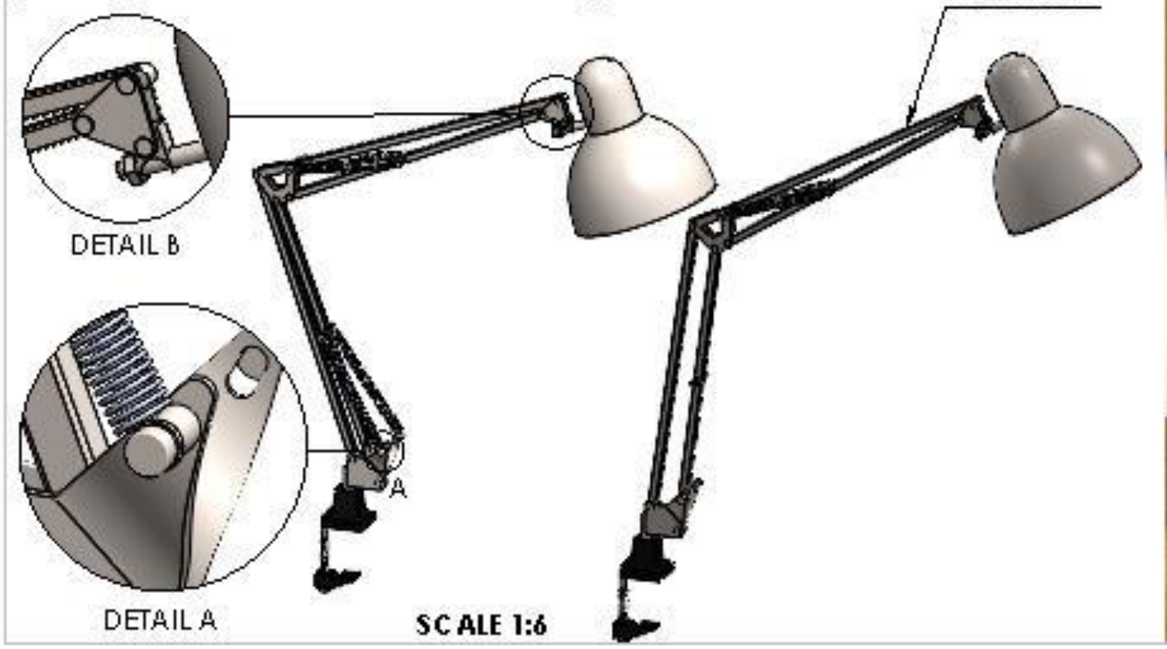


SCALE 1:4

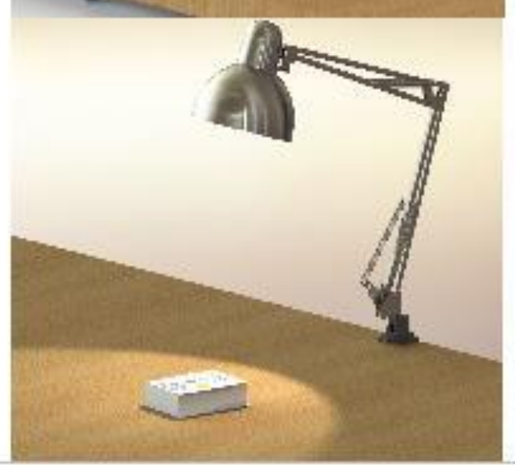
PLAN



MOVEMENT

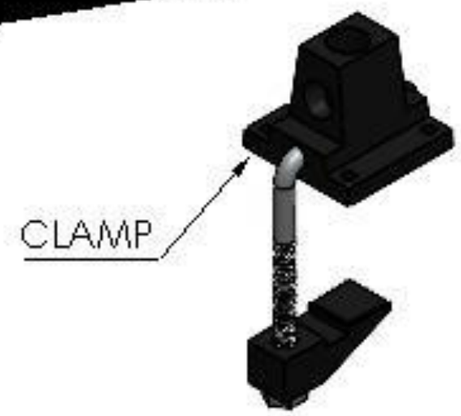


SCALE 1:6



SPRING

CLAMP



DESIGN AND COMMUNICATION GRAPHICS

PART A: OUTPUTS

Desk Light

CASTLEROY COLLEGE

EXAM NUMBER: 141716

Desk Light

- Exploration Of Concept design

- Theme: ROBOTICS

WHY?

I chose this theme of Robotics because i've always had a **keen interest** in **technology** and the overall advancement of it. Robotics is naturally associated with the theme of artificial intelligence and over the years the evolution of what is depicted as a robot has changed. It **amazes me** how **sophisticated** and **complex** a **robot can be** and it does not specifically have to be human like, however the way we first perceived the idea of a human like robot is drastically different to the day we live in today and it **goes to show the achievements that we have obtained through the study of robotics.** This is why i would like to incorporate the theme of robotics into the exploration of my concept design. Robotics is **constantly in the mouth of people today** and i believe it be **very important** because it will definitely **define our future** which makes this theme a suitable choice.

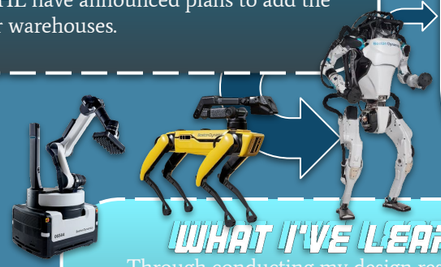
TARGET MARKET

The target market for my design will be **those** who are **interested** in the **field** of **science** and **technology** . This may include **simple tech enthusiasts** , **gamers** , **students** , **younger children** or even **educational institutions** that may want a robotic themed desk light to make their technology sector more engaging. I hope to design a desk light that appeals to this target market to **share my keen interest** on this topic. I also hope to make this desk light aesthetically pleasing to suit the environment it may be set in and not create an eye sore, the desk light must be practical as well. It should have **simple functionality** and **nothing too complex** as the target market will also suit to **younger children** who may have **technology themed rooms.**

BOSTON DYNAMICS

Boston dynamics is currently one of the **world's leading companies** in terms of **developing and deploying highly mobile robots.** They're responsible for the famous creations of the **"Spot Robot** ", "**Atlas** " and "**Stretch** ". The most important of the 3 in the factory industry would be Stretch. **Stretch is a multi-purpose mobile robot designed to support material handling tasks** , such as truck and container unloading. Since its release in 2021 companies such as DHL have announced plans to add the robot in their warehouses.

Boston dynamics have has also **inspired me** to do this theme because of their breakthrough of the Spot and Atlas robot. Typically it would've been thought that you'd only see such robots in movies but they brought it to life.



WHAT I'VE LEARNED

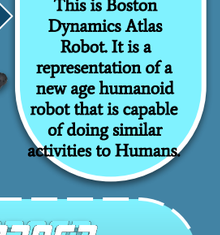
Through conducting my design research, I have discovered lots of information about desk lights. I have learned a significant amount about the **practicality of varying light levels** , how the **temperature from a bulb affects the atmosphere** and the **true importance of having an adjustable desk light.** I hope to use this information when designing my own model as well. I have **identified** these **issues** and **acknowledged strengths** in the various desk lights I've researched, especially the two I did my comparison on. From this, I am now sure of the features I would like to include in my concept design.

Examples of robots in the industry:

These industrial robots have revolutionised the manufacturing of cars making the process a lot faster and less labour extensive for humans.

This bomb disposal robot has helped saved countless lives. It's manufacturing prevents actual people from needing to be close to defuse the bomb.

Nasa's Rovers is also a breakthrough in Space exploration. This robot "perseverance" landed on mars in search of ancient life.



This is Boston Dynamics Atlas Robot. It is a representation of a new age humanoid robot that is capable of doing similar activities to Humans.

ROBOTICS

Robotics is in the **field** of **science** and **technology** and it **focuses** on **creating** and **designing machines** called robots . These robots / machines can **perform tasks** or **actions on their own** or often with the help of computer programs. Robots come in **all shapes** and **sizes** , from small ones used for cleaning floors to big ones used in factories. They can do things like moving, grabbing objects, and even making decisions based on what they see or sense. Robotics is all about **making smart machines that are capable of helping people with various tasks.** This can include everyday chores all the way to complex jobs in industries and exploration. Robotics is an **ever growing field** and continues to **evolve rapidly. Breakthroughs** in robotics have the **potential to significantly impact various aspects of our lives and society** as a whole and this is already being seen throughout the world today.

COLOUR SCHEME

The **Main colours associated** with robotics are the standard picks of black , grey **and white.** These colours are the **most natural** and **easiest** to **source** which can **reduce cost** of material as the **mechanics behind robotics already cost a significant amount.** A frequent use of **metal** is also used in robotics which makes **silver/grey** a widely used colour whilst also making the robot look **high-tech and futuristic.** However, people may deviate from silver and gray but will still commonly use a **metallic finished** colour. **Accent colours** may include red , yellow and blue . These colours are often seen in the **lights emitted by robots.** Red and yellow most **often depict caution.** They are often seen on specific parts of robots especially on **industrial or safety robots.** Blue can suggest **trust, reliability, and a friendly demeanor.** It's often used for **robots** designed to **interact** with **people** , such as **social** robots or **robotic toy**

Main Colours	Accent colours

EXISTING PROBLEMS

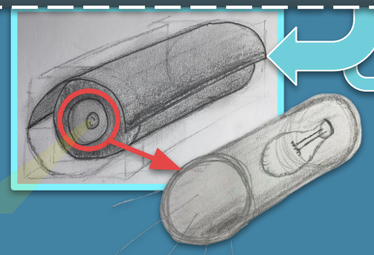
From my design research i noticed that a lot of existing desk lights still have not made the transfer to **led light bulbs.** While this has improved, there is still a substantial amount of desk lights on the market that are still using **filament bulbs** . Due to this, sustainability and efficiency in desk lights are affected as well as safety because of the filament light bulbs ability to get very hot. As a result, i intend to incorporate the use of an led light in my design.

Desk Light
- Shape and Form

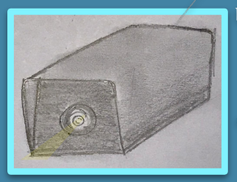
Having Looked at robotics, especially **boston dynamics** atlas robot, i took notice at the **humanoid structure** . This led me to the conclusion that i wanted to create something similar but have it more **friendly to the eye**

LAMP SHADE

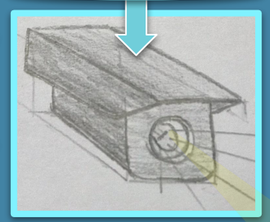
Whilst Looking at the **boston dynamic atlas robot** , i took special notice at the **camera on its head** . This observation led me to the thought about all the different types of **cameras** that may be associated with robotics.

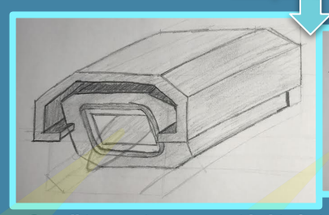


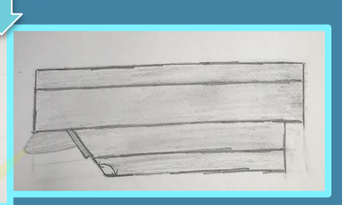
I Initially liked the idea of having a cylindrical based camera as the lamp shade. It resembled the shape of **an eye** with an **eye brow** which relates to a **humanoid robot**. It also has the likes of a cctv camera which allowed me delve further into cameras and gain more inspiration.



Instead of a circular based camera, i tried for a camera with a more rigid rectangular/square structure to more closely portray the likes of a **head.** I much preferred this shape especially with the sketch on the right so i attempted to refine it.



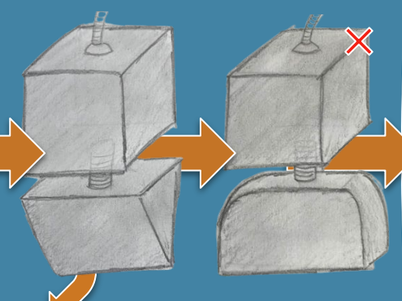




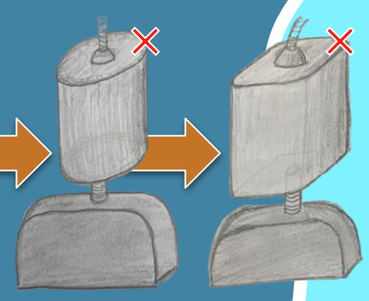
Overall i was quite content with this design. I decided to add an **angle** to which the face of the light would point. I think this helps emphasise the eye brow on top and also makes it easier to focus illumination on objects right below the desk light. The design is even more rigid than before and i think it has the structure i want to be implemented onto my final solution.

MAIN BODY

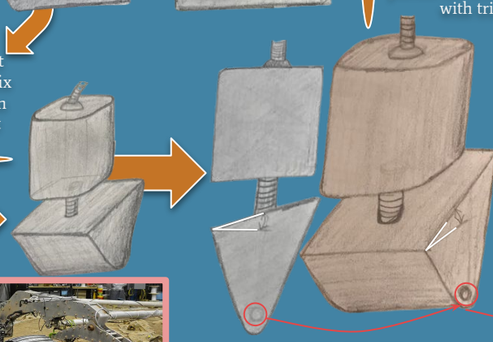
When thinking about what do for the main body i came to the conclusion that it shouldn't closely resemble the anatomy of a human body. Although some may like the idea, i believe it wouldn't be as aesthetically pleasing to have. For this reason i have opted for a more retro robot humanoid look with more a more boxy shaping instead



I first started off with the overall shape. The main body needed a torso and hip shaped bottom half to get the main structure of a body. I felt that the triangular shaped hip looked better as opposed to the filleted square. I thought that the reason for this might have been the fact the torso was not filleted or cylindrical so i proceeded to sketch it as a cylindrical shape but i was still not content with the look. This led me to pursue the template of the cuboid torso with triangular hip shape.



I liked the form of this design but not as much the harsh edges. To fix this i decided to fillet the edges on the cuboid and the bottom point of the hip.



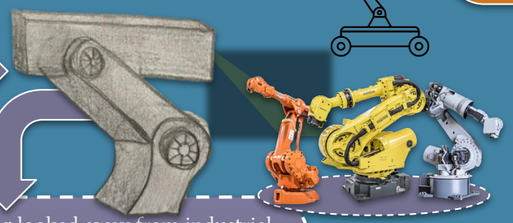
In addition to the fillets i also decided to give the triangular hip a tilt to make it look like its moving forward. I think it's quite a subtle change but i do also feel it adds to the overall structure and look to the body,

BASE

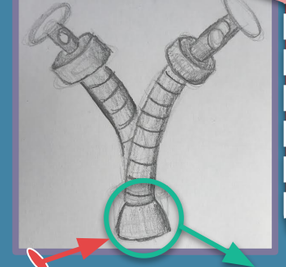
With regard to the holes in the main body i figured the **base** could be **connected** to the **main body through** these **holes** , similar to a **hip joint** . Since i already had features similar to the likes of a head, neck, torso and hip, i needed to design something that would also **mimic the use of legs** . For this reason i looked towards **wheels** because i liked the idea of it being **portable** and to move around on the desk

MOVEABLE NECK

Having researched industrial robots i learned that their necks are moved mainly through **articulated arms** as seen on boston dynamics "Stretch" and other industrial robots.



However, having looked away from industrial robots and to actual humans i figured that a **goosenecked structure** might be **more viable** as it closely resembles how our own necks move. Due to the fact that i want to incorporate the likes of humanoid robots into my final solution, i believe that the gooseneck design is most suitable



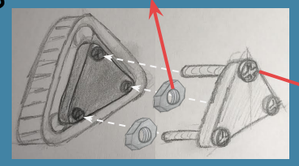
Instead of the simple cylindrical piece connecting the goose neck to the body, i tried a chamfered cylindrical piece instead. Although it was different i much preferred if it was a more seamless transition.

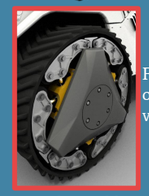
In order to achieve a more seamless look i opted for an asymmetric filet on the cylindrical piece. I believe this to look much better than the standard cylindrical piece and chamfered piece as it tidies up the harsh edges.

I much preferred this look and do believe that it makes more sense for my final solution. So i proceeded to refine it and add more detail to the design

Face plate should be a steel/strong metal to strengthen the structure of the triangular caterpillar track wheel

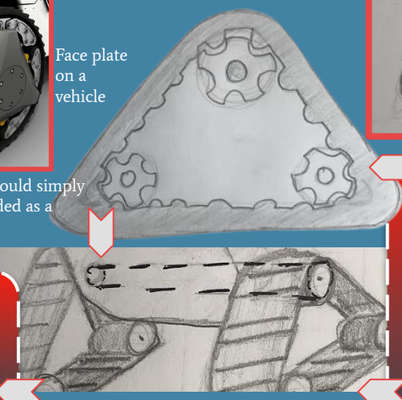
Hex nut to tighten face plate against gear





Face plate on a vehicle

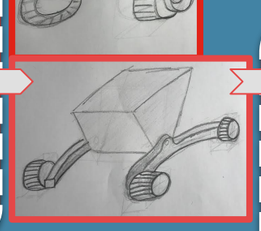
Axle would simply be binded as a screw



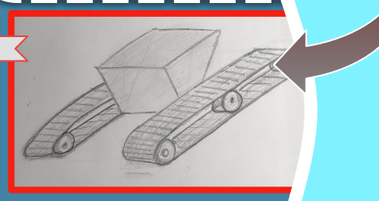
In order to connect the top axle to the bottom gears to hold the caterpillar tracks structure, I implemented a face plate. In my opinion having this face plate makes piece look better. It hides the machinery and cogs of the moving caterpillar tracks making the base more sleek

To start out i sketched a pair of wheels on each side. I liked this idea but i decided to sketch other ideas before choosing it.

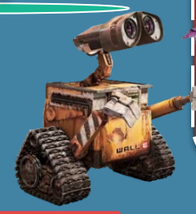
For my next sketch i took influence from the wheels of a space rover. I also really liked this design as well as it was much more unique than a normal set of wheels



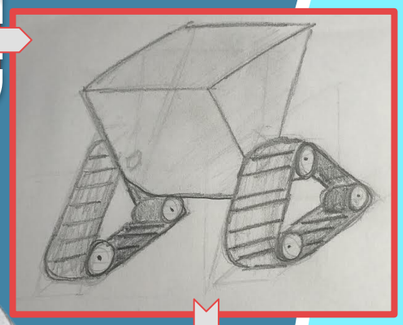
Although i really liked the idea of wheels, i had faced the realisation that they might **move/roll off surfaces too easily** which would often remove the focussed illumination on the desired spot. I built off this realisation and proceeded to think about something that would provide a **more stable from.** This led me to the conclusion of **caterpillar track wheels** as they had a **large flat surface area** against the surface they were on.



I liked this idea but i felt that this type of caterpillar track looked wrong. My desk light will be vertically tall, and having a long perpendicular base will make it look unproportionate.



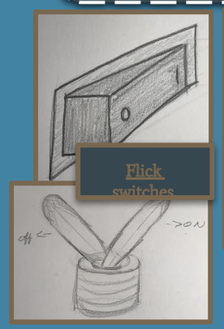
researched further into robots with tracks and i was taken by the famous robot "Wall-E" and his triangular shaped caterpillar track wheels.

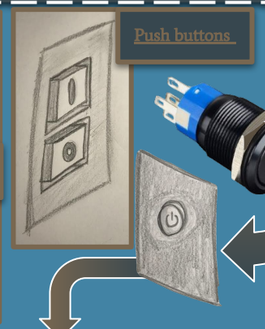


I was much more content with this design. The wheels weren't too long and i think it contrasted well with the triangular shaped body, it was much more proportionally well. With the shape and form of the tracks achieved, i now had to figure out how they were attached to the body and how the tracks operated. I solved this by a simple gear intertwining with the grooves from the track and inserting an axle through the main body and into the gear.

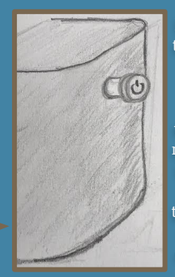
CONTROLS

The physical controls were quite easy to think about as these are the things that turn on essentially everything in our environment. Some ideas i had came up with were sketched as shown.





Although thinking about these buttons were easy, choosing one was the most difficult part.
I liked the push to make buttons as it resembled a sort of core. I felt having one button near to where the heart would be would give it a more humanoid characteristic so i stuck with it.

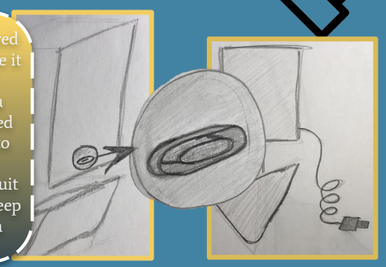


Instead of having the button almost flush with the body, i drew it coming out. I think this created more depth to the body and further highlighting the the significance of the core turning on the desk light.

After this sketch i realised it looked like the stop buttons on the machinery you would see in an industrial factory so i thought it would be a good idea to give it that red textured eye catching finish.

WIRELESS

Since the robot was on wheels i figured it would be a great advantage to make it wireless. This added to its ease of portability. It would be charged via USBC on the back. If the user wanted didn't want to worry about having to charge it they could simply leave charger plugged in. The internal circuit would trickle charge the battery to keep it at full capacity much similar to a smartphone.



FINALISATION OF DESIGN

COST

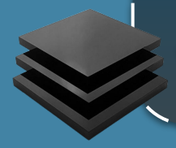
As the target audience is quite broad, i want the desk light to be fairly priced to encapsulate the whole audience allowing everyone to purchase it. The factor mainly influencing the cost will be the material of the desk light. Another article to note is the **extra functionality** of the desk light, this being the app and the adjustment of the light bulb.

material

app

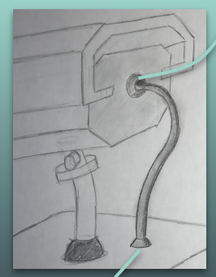
light bulb

MATERIAL



The material of the desk light will mainly be plastic. The plastic should be **strong, tough** and **durable** able to withstand any unanticipated stresses or impacts. With regard to the **bottom base** of the desk light and it's wheels, It should also be durable. The **face plate** binding the gears should be made of a **stainless steel,** the **screws as well** . The **tracks** themselves will be made from **rubber** and this is where most of the weight for the desk light will go in order to keep the desk light base heavy.

CABLE



Also asymmetric connector. (insulator material)

Instead of keeping the **wire** from the battery **internally** through the goose neck and to the bulb, i put it outside. **This idea was mainly for aesthetic purpose.** I believe that by showing some wiring, but not alot would make help the robotic aesthetic more prominent.

Also asymmetric connector. (insulator material)

APP

I think it would be a great addition to have an **app** for the adjustment of the desk light **rather than** physical controls or a remote. As the **desk light has the potential to be wireless** i think that having an app would be a great extra **quality of life feature** being able to **control it from a distance.**

When purchased and **first set up** the desk light will **automatically be fixed at a warm light temperature.** As you **setup the app** and connect to the desk light you will be able to **change** light bulb **brightness** , **temperatures** , **colours** and **different ambience lighting modes.** On top of this, the app will serve as a **communication platform** between the consumer and staff to help with any inquiries they may have.

GET IT ON Google Play

Download on the App Store

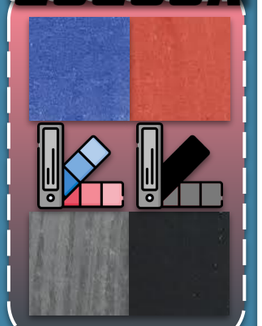
The app will be available on the App Store and Google Play store.
You will receive an access code when purchasing the desk light allowing you to use the app

On top of this i'd like the app to have a **designated section** providing **relevant information** on anything in the **technology** industry helping the consumer stay up to date on the latest tech ne

LIGHTBULB

As **filament bulbs** are an **existing issue** with some modern day devices i have ensured that the **led bulb** was used, the rgb customization is only a greater perk to using these types of bulbs and it will be hugely beneficial especially if this desk light were to sell in mass quantity as it reduces the amount of hazardous material spread.

COLOUR



As previously mentioned the colour blue suggested **trust** and **reliability** with a friendly demeanor. With that, i have decided to give the robot the **primary colour** of blue on the **lamp shade** and **main body**. The red would be placed on the **on/off switch** to catch the consumers attention. The **black** would be associated with the **rubber track wheels** and the grey would represent the **steel colour** of the **faceplates** and the **goosenecks** of the desk light.

SUSTAINABILITY

With regard to the plastic being used i would also like to make it from completely **recycled plastics.** Since the main body and lamp shade are plastic heavy it would be hugely unsustainable if it were not made from recycled plastic.

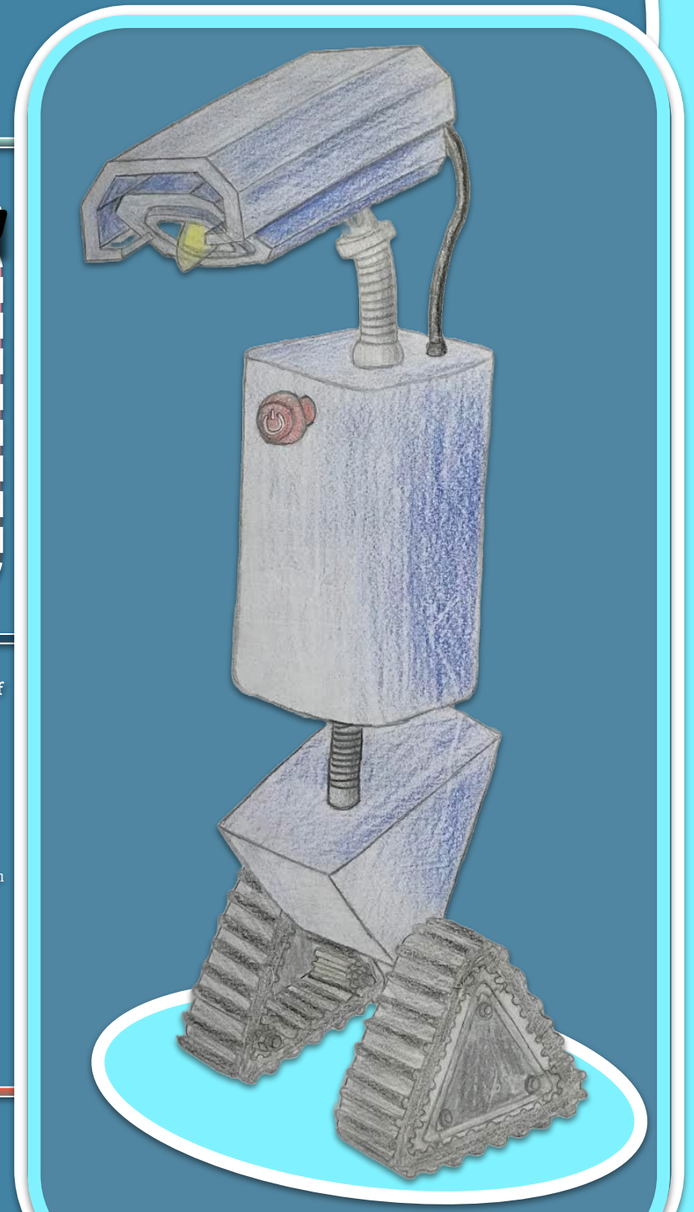
WEIGHT

The **weight** of the desk light will mostly be **present in the base** . This will allow the desk light to **stay upright** consistently without falling. I hope to aim for around **0.5-1 kg** for the base of the desk light to ensure this happens. With reference to everything **above** , it should be **lighter than** the **weight on** the **base**. This will help **prevent** the desk light from **toppling over unsuspectedly** or drooping over.

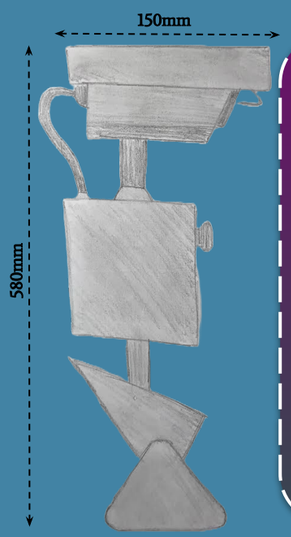
TEXTURE

For the main body, on/off switch and **cable** i have decided to go with a **high gloss finish on the plastic.** I believe this would enhance the appearance of the desk light as it's reflective nature adds depth and richness to its colour. With regard to the cable, the black high gloss would represent a simple pvc insulating material.

For the wheels i opted for the **tire finish** as the **tracks** were rubberised and **stainless steel** would then be used for the **face plates.**



SIZE



Desk lights can vary in size drastically. For this reason there is no optimal size for a desk light but just big enough fit onto a desk. I want to have the **height** of the desk light **relatively large** so it may be **long enough** to **stretch** into certain areas to **compromise** for the **fixed extendability** of the **gooseneck** . The height should not exceed no more than **600mm** vertically and **200mm** horizontally

ABB

FANUC

SAFETY

To ensure safety amongst the use of desk lights standard safety features must be implemented into the concept design. In the rare instance that the desk light gets **too hot** , **thermal sensors** must be installed to detect this heat and then **turn off** the desk light to **stop it** from **overheating** and **breaking** , potentially causing harm or even fire

FUNCTIONALITY OF FINAL DESIGN

My final desk light design will **operate efficiently** as sought fit. It features **clear maneuverability** in two regions, the top and bottom via **gooseneck design** . It will be **wireless** and can move on surfaces due to its **triangular caterpillar track wheels** . The cctv camera will be tilted at an **angle** which serves to emit light at a downward angle onto the desk. The light will be emitted by an **rgb led bulb** that can be controlled **wirelessly** by a **coherent user friendly app**

REFLECTION

Overall i am quite pleased with how my design turned out. I was influenced by boston dynamics humanoid robot and wanted to create my own and the the product achieved at the end completely suits to my liking. I believe that it fits into its target audience previously stated. The robot is unique and has not yet been done before, the shapes and colours have been heavily influenced by the theme and i'm very interested to see how it will progress onto solidworks.

FREEHAND GRAPHICAL REPRESENTATION

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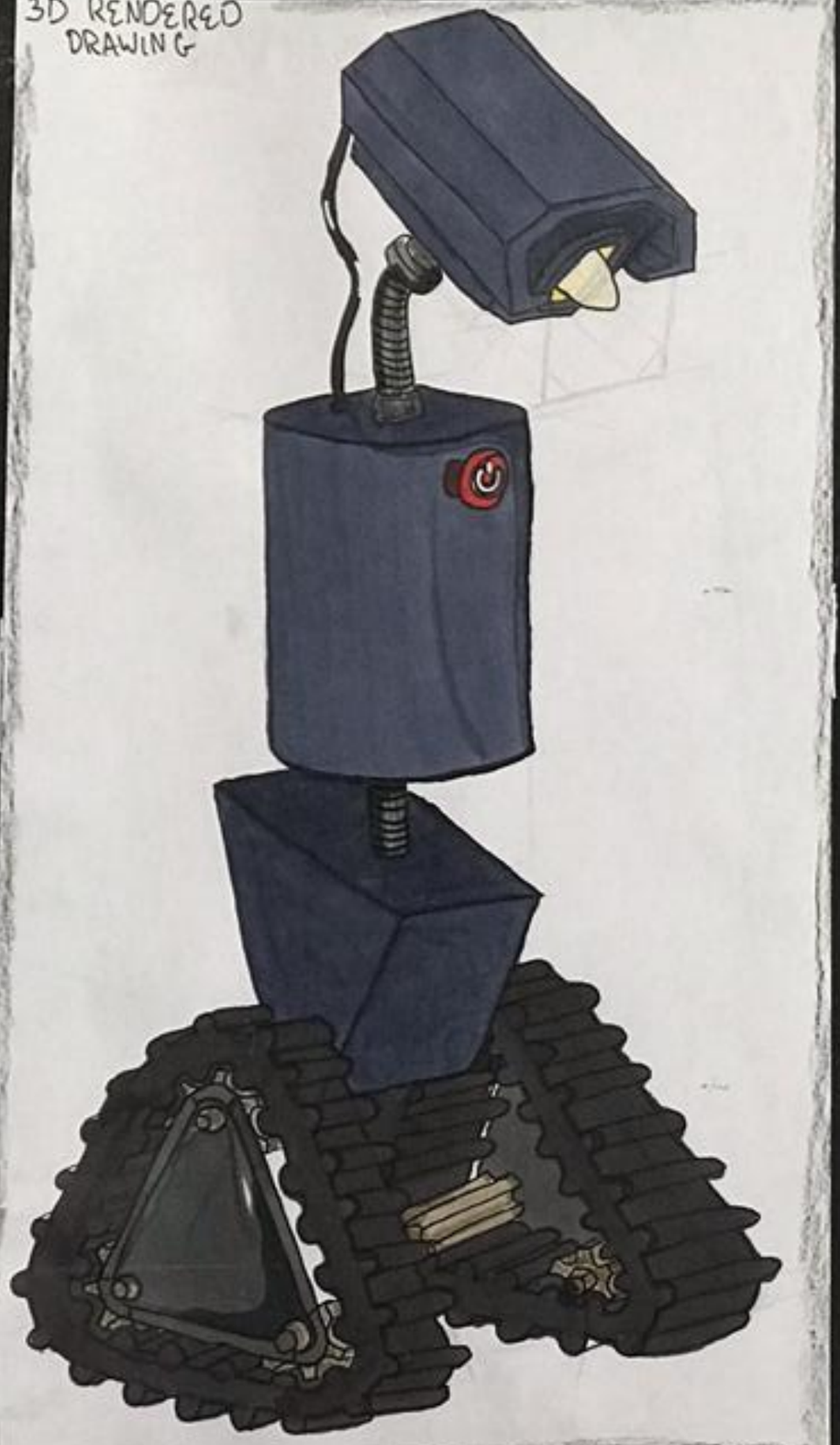
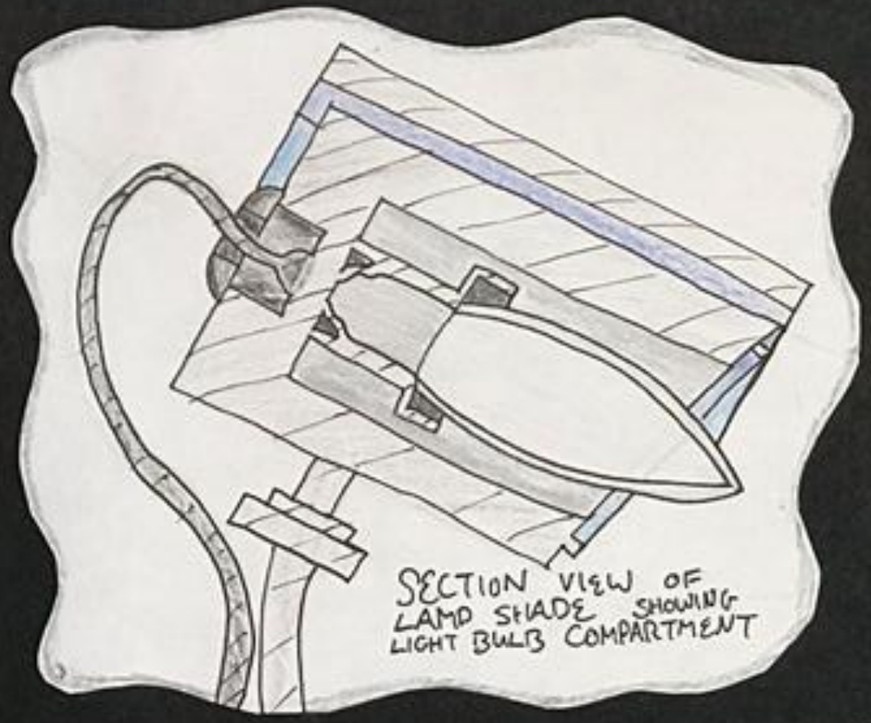
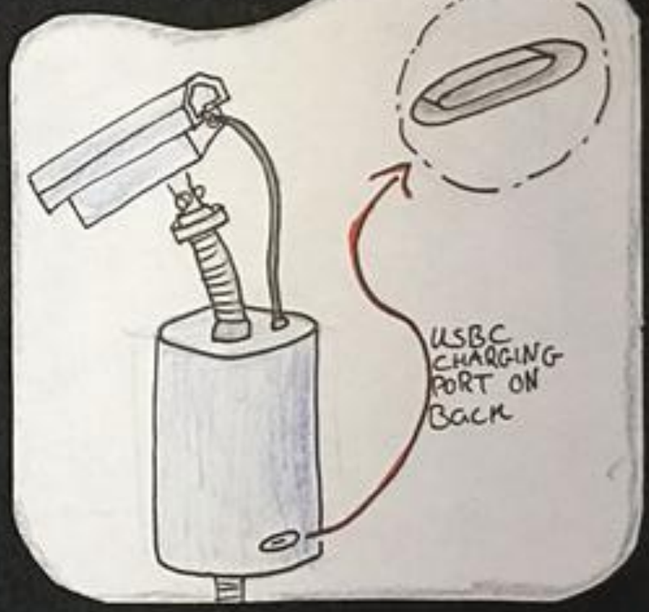
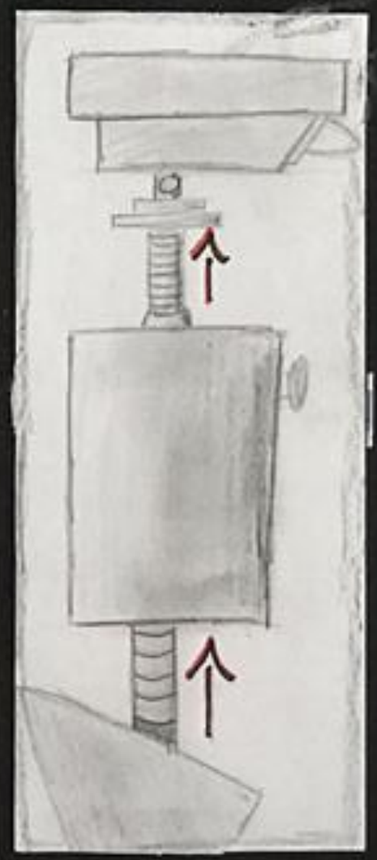
3D RENDERED DRAWING



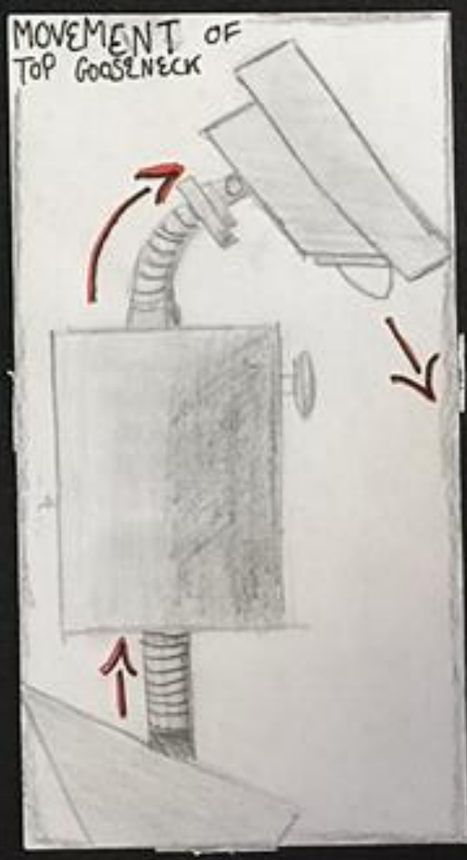


SECTION VIEW OF LAMP SHADE SHOWING LIGHT BULB COMPARTMENT



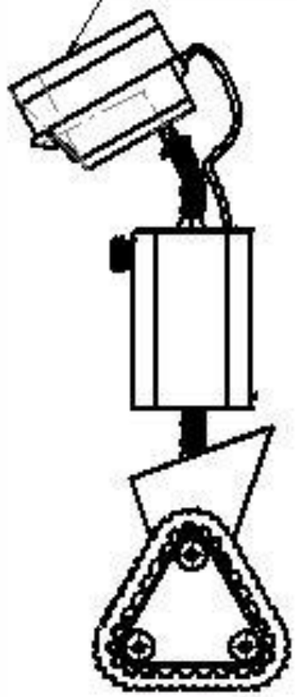


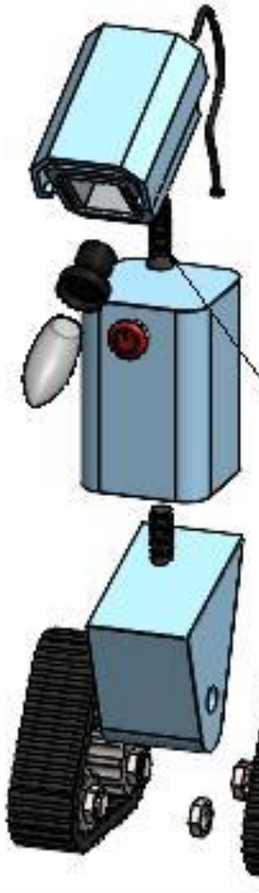
MOVEMENT OF TOP GOOSENECK



MOVEMENT OF TOP & BOTTOM GOOSENECK

END ELEVATION

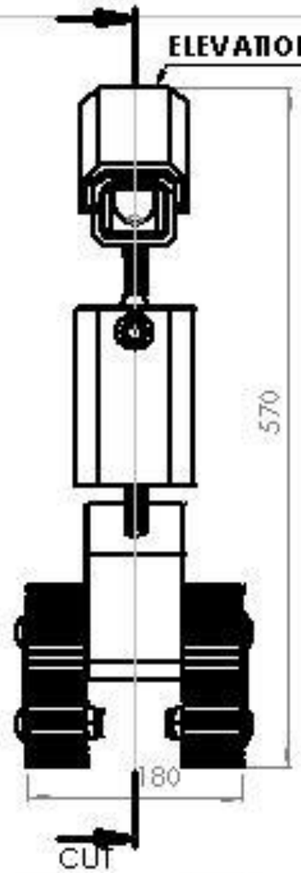




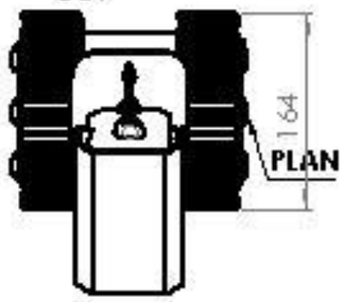
EXPLODED VIEW

CUT

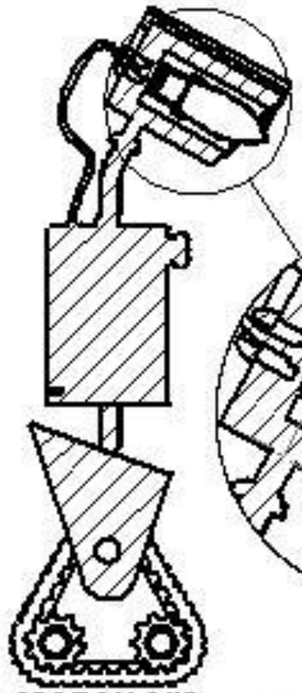
ELEVATION



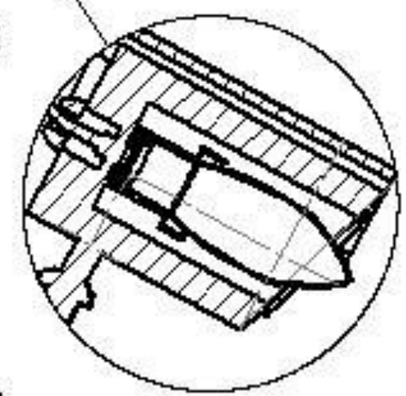
CUT



PLAN

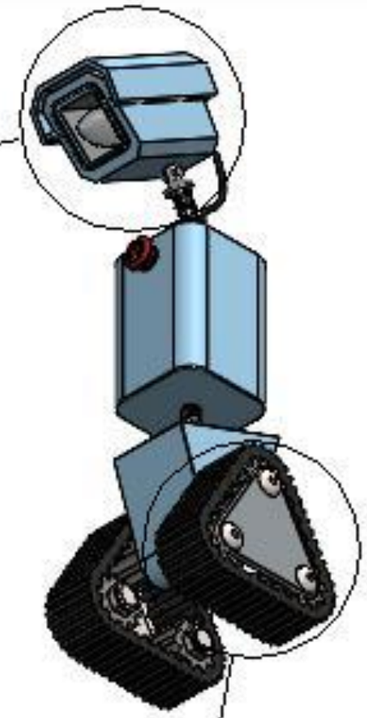


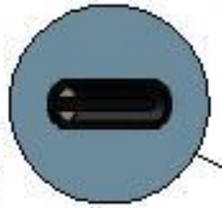
SECTION CUT



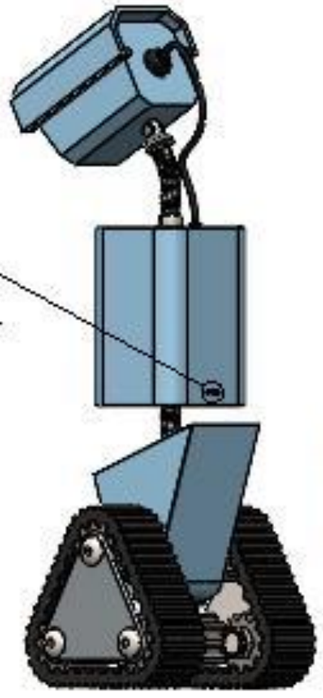
LIGHTBULB
SCALE 2 : 5

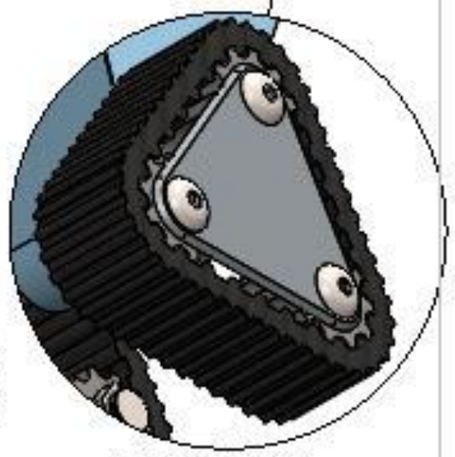
CAMERA LAMP SHADE
SCALE 2 : 5



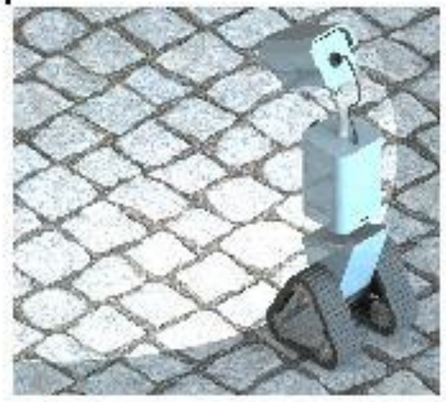


CHARGING PORT





TRACK WHEELS
SCALE 2 : 5



DESIGN AND COMMUNICATION GRAPHICS

PART B: OUTPUT 7

Desk Light

CASTLEROY COLLEGE

EXAM NUMBER: 141716

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Race car desk light
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Color temperature:
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